

Trace vanishing of $\Omega^{(\lambda)}$ on V^λ from the multi-row meta-theorem

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Abstract

Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ with $\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|) \geq 1$. We prove $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)}) = 0$ as a direct corollary of the 2026-05-15 multi-row meta-theorem (Pillar 1: image-in- E_1^-), combined with a one-line algebraic identity in the Iwahori–Hecke algebra: $\Omega^{(\lambda)}(T_1 + 1) = (q + 1)\Omega^{(\lambda)}$, which follows from $(T_1 + 1)^2 = (q + 1)(T_1 + 1)$ and the fact that $\Omega^{(\lambda)}$ ends in the rightmost factor $(T_1 + 1)$ of R'_n . Trace-vanishing is then a one-line corollary of the proved meta-theorem. We also note the stronger structural corollary: in the Hoefsmit seminormal basis ordered by descent at position 1, the matrix of $\Omega^{(\lambda)}$ has block form $\begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$, in particular $\Omega^{(\lambda)2} = 0$ on V^λ .

1 Setup

We follow the conventions of the May-15 multi-row meta-paper. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q - 1)T_j + q$. The eigenvalues of each T_j on any module are therefore q and -1 . Set $S_j := T_j + 1$. For $k \geq 1$,

$$R'_k := S_{k-1}S_{k-2} \cdots S_1 = (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_1 + 1) \quad (R'_1 := 1).$$

For $\nu \vdash N$ with $\ell = \ell(\nu)$, set

$$\Omega^{(\nu)} := R'^{(\nu)}_{\ell+1} R'^{(\nu)}_{\ell+2} \cdots R'^{(\nu)}_N.$$

Throughout this paper, $\widehat{\lambda}$ is a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row of length ≥ 2 . We set

$$\lambda := (\widehat{\lambda}_1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})}, \underbrace{1, \dots, 1}_q) \vdash n, \quad n = q + |\widehat{\lambda}|, \quad \ell := \ell(\lambda) = \ell(\widehat{\lambda}) + q,$$

and the threshold

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|).$$

V^λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$. For each $j \in \{1, \dots, n-1\}$, write

$$E_j^+ := \ker(T_j - q)|_{V^\lambda}, \quad E_j^- := \ker(T_j + 1)|_{V^\lambda}.$$

Descent at position 1. For $T \in \text{SYT}(\lambda)$, entry 1 sits at cell $(1, 1)$ and entry 2 sits at one of the cells $(1, 2)$ or $(2, 1)$. We say T is *descent-0 at position 1* if 2 is in row 1 of T and *descent-1 at position 1* if 2 is in column 1 of T . The axial distance between cells of 1 and 2 is then $+1$ or -1 respectively, so

$$T_1 v_T = \begin{cases} q v_T, & T \text{ descent-0} \\ -v_T, & T \text{ descent-1.} \end{cases} \quad (1)$$

In particular T_1 acts diagonally on the Hoefsmit basis, and

$$E_1^+|_{V^\lambda} = \text{span}\{v_T : T \text{ descent-0}\}, \quad E_1^-|_{V^\lambda} = \text{span}\{v_T : T \text{ descent-1}\}. \quad (2)$$

2 Inputs

Theorem 1 (Pillar 1: multi-row meta-theorem). ([?, Theorem 1].) *For $\widehat{\lambda}$ as above and $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ (equivalently $\tau(\widehat{\lambda}) \geq 1$),*

$$\Omega^{(\lambda)} V^\lambda \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V^\lambda}.$$

In particular, $\Omega^{(\lambda)} V^\lambda \subseteq E_1^-|_{V^\lambda}$.

Lemma 2 (Right-absorption identity). *As an element of $H_q(S_n)$,*

$$\Omega^{(\lambda)} (T_1 + 1) = (q + 1) \Omega^{(\lambda)}.$$

Proof. The Hecke quadratic relation $T_1^2 = (q - 1)T_1 + q$ gives

$$(T_1 + 1)^2 = T_1^2 + 2T_1 + 1 = (q + 1)T_1 + (q + 1) = (q + 1)(T_1 + 1).$$

The factor $R'_n = (T_{n-1} + 1)(T_{n-2} + 1) \cdots (T_1 + 1)$ ends with $(T_1 + 1)$ on the right. Therefore

$$R'_n \cdot (T_1 + 1) = (T_{n-1} + 1) \cdots (T_2 + 1) \cdot (T_1 + 1)^2 = (q + 1) R'_n.$$

Multiplying on the left by $R'_{\ell+1} \cdots R'_{n-1}$ and pulling the scalar through gives $\Omega^{(\lambda)}(T_1 + 1) = (q + 1)\Omega^{(\lambda)}$. $\square$

Remark 3. Lemma ?? is an identity in $H_q(S_n)$, independent of λ : it depends only on the rightmost factor of $\Omega^{(\lambda)}$ being $(T_1 + 1)$. In particular it implies $\Omega^{(\lambda)} \cdot E_1^- = 0$ on *every* H_q -module: for $v \in E_1^-$ we have $(T_1 + 1)v = 0$, so $(q + 1)\Omega v = \Omega(T_1 + 1)v = 0$, hence $\Omega v = 0$ ($q + 1$ is a unit in $\mathbb{Q}(q)$). This is the structurally trivial right-vanishing on E_1^- , holding for every shape.

3 Main theorem

Theorem 4 (Trace vanishing on V^λ). *Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ and every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ with $\tau(\widehat{\lambda}) \geq 1$. Then*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) = 0.$$

Proof. Compute the trace in the Hoefsmit seminormal basis:

$$\mathrm{tr}_{V^\lambda}(\Omega) = \sum_{T \in \mathrm{SYT}(\lambda)} [v_T](\Omega v_T),$$

where $[v_T](w)$ denotes the coefficient of v_T in the expansion of w in the seminormal basis. We split the sum by descent at position 1.

Case D0: T descent-0. By Theorem ??, $\Omega v_T \in E_1^-|_{V^\lambda}$. By (??), $E_1^-|_{V^\lambda}$ is spanned by descent-1 basis vectors. By linear independence of the Hoefsmit basis, the coefficient of the descent-0 vector v_T in any vector of $E_1^-|_{V^\lambda}$ is zero. Hence $[v_T](\Omega v_T) = 0$.

Case D1: T descent-1. By (??), $(T_1 + 1)v_T = 0$. By Lemma ?? as operators on V^λ ,

$$0 = \Omega(T_1 + 1)v_T = (q + 1)\Omega v_T.$$

Since $q + 1$ is a unit in $\mathbb{Q}(q)$, $\Omega v_T = 0$. In particular $[v_T](\Omega v_T) = 0$.

The trace vanishes term by term. $\square$

Corollary 5 (Block-strict-lower form). *Order the Hoefsmit basis of V^λ as $(v_T : T \text{ descent-0} ; v_T : T \text{ descent-1})$, and write $M(\Omega)$ for the matrix of $\Omega^{(\lambda)}$ in this basis. Under the hypotheses of Theorem ??,*

$$M(\Omega) = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$$

for some matrix X . Consequently $\Omega^2 = 0$ on V^λ , $\mathrm{rank}(\Omega^{(\lambda)}|_{V^\lambda}) = \mathrm{rank}(X) \leq \min(d_0, d_1)$ where $d_i = |\{T \in \mathrm{SYT}(\lambda) : T \text{ descent-}i\}|$, and $\mathrm{tr}(\Omega) = 0$.

Proof. For T descent-0, $\Omega v_T \in E_1^-|_{V^\lambda}$ is supported only on descent-1 basis vectors (Theorem ?? + (??)); column T of $M(\Omega)$ has zeros in descent-0 rows. For T descent-1, $\Omega v_T = 0$ (Lemma ?? applied to $v_T \in E_1^-$); column T is identically zero. This gives the stated block form. Then $M(\Omega)^2 = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix} = 0$, and $\mathrm{tr} M(\Omega) = 0 + 0 = 0$. $\square$

4 Computational verification

The forward-direction trace vanishing was verified empirically (prior to this proof) on six shapes via symbolic computation in `sympy` with q symbolic. The two $\tau \geq 1$ shapes within the multi-row meta scope are:

λ	$\ell(\widehat{\lambda})$	$ \widehat{\lambda} $	q	τ	$\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)})$
$(3, 2, 1, 1, 1)$	2	5	3	1	0
$(3, 2, 1, 1, 1, 1)$	2	5	4	2	0

In addition, four $\tau = 0$ shapes $((3, 2, 1), (3, 2, 1, 1), (4, 2, 1), (3, 3, 1))$ were tested and have non-zero trace, consistent with the forward direction (and providing evidence for the still-open converse).

The script `~/projects/scratch/2026-05-16-trace-vanishing-strategy/two_sided_signkill_test.py` also independently verifies the structural Corollary ??: the matrix of $\Omega^{(\lambda)}$ on $V^{(3,2,1,1,1)}$ in the Hoefsmit basis has the block-strict-lower form, with rank 2.

5 Open: converse direction

The converse direction

$$\tau(\widehat{\lambda}) = 0 \implies \mathrm{tr}_{V^{\lambda}}(\Omega^{(\lambda)}) \neq 0$$

remains open. Empirical evidence: four $\tau = 0$ shapes verified non-zero (above). The proof in this paper does not address the converse; it gives only the forward direction. A natural next step is leading-coefficient analysis of $\mathrm{tr}(\Omega)$ as a polynomial in q , or an explicit non-vanishing argument from the seminormal-basis matrix at descent-0 diagonal entries.

6 Methodological note

This proof is a near-immediate corollary of the multi-row meta-theorem plus the trivial right-absorption identity. The discovery process (2026-05-15 wake-2 through 2026-05-16 wake afternoon) is recorded in the dream journal and need not concern us here, except for one methodological observation: when a new conjecture appears, check first whether it follows from existing pillars and an elementary algebraic identity, before invoking heavy machinery (Markov traces, JM products, categorical lifts). In the present case, the trace-vanishing conjecture admitted a one-paragraph proof from already-proved Pillar 1 and a one-line lemma; the more elaborate attack routes (q-binomial factorisation, JM-product involution, Markov-trace cancellation) were not needed for the forward direction.

References

- [1] C. *Extended sign-kill for general multi-row $\widehat{\lambda}$: the meta-theorem at arbitrary $\ell(\widehat{\lambda}) \geq 2$.* ~/projects/proofs/2026-05-15-multi-row-sign-kill-meta.tex (commit 7936707 on clio-vega/proofs, 2026-05-15.)